UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 12/09/2024*

PROBLEM

We observe N samples of the following discrete-time real random process:

$$X[n] = A \cdot s[n] + W[n], \quad n = 0, 1, \cdots, N-1,$$

where A is the amplitude of the Signal of Interest (SOI) and it is modelled as a deterministic unknown parameter, $s[n] = (-1)^n$ is the signal shape, perfectly known, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) given by $S_W\left(e^{j2\pi f}\right) = \sigma^2$.

(1) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq \left[X[0] \quad X[1] \quad \cdots \quad X[N-1]\right]^T$; the pdf can be expressed either in scalar form and in vector form, **provide both**.

(2) Under the assumption that σ^2 is **a-priori known**, derive the Maximum Likelihood (ML) estimator of the amplitude A. Derive the pdf of the estimator $\hat{A}_{ML}$ and then its bias and mean square error (MSE). Finally, derive the Cramér-Rao lower bound (CRB) on the estimate of A and compare it with the MSE of the ML estimator; establish whether the ML estimator is biased, consistent, and efficient.

(3) Under the assumption that σ^2 is also **unknown**, derive the joint ML estimator of signal amplitude A and noise power σ^2. Determine whether the two ML estimators are biased or not.

Solution

(1) The joint pdf of the observed data vector is:

$X[n] \in N\left(As[n], \sigma^2\right), \quad \left\{X[n]\right\}_{n=0}^{N-1}$ indipendenti $\Rightarrow \quad \mathbf{X} \in N\left(A\mathbf{s}, \sigma^2 \mathbf{I}\right)$, where $\left[\mathbf{s}\right]_n = s[n] = (-1)^n$.

The pdf in scalar format is:

$$f_X(\mathbf{x}) = \prod_{n=0}^{N-1} f_X(x[n]) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left(-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\left(x[n] - A(-1)^n\right)^2\right)$$

The pdf in vector format is:

$$f_X(\mathbf{x}) = \frac{1}{\left(2\pi\right)^{N/2}\sqrt{\left|\sigma^2\mathbf{I}\right|}} \exp\left(-\frac{1}{2}\left(\mathbf{x} - A\mathbf{s}\right)^T \left(\sigma^2\mathbf{I}\right)^{-1}\left(\mathbf{x} - A\mathbf{s}\right)\right) = \frac{1}{\left(2\pi\right)^{N/2}\left(\sigma^2\right)^{N/2}} \exp\left(-\frac{1}{2\sigma^2}\left\|\mathbf{x} - A\mathbf{s}\right\|^2\right)$$

(2) Under the assumption that σ^2 is a-priori known, the log-likelihood function is:

$$\ln L(A) = \ln f_X(\mathbf{x}) = -\frac{N}{2}\ln\left(2\pi\right) - \frac{N}{2}\ln\left(\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\left(x[n] - A(-1)^n\right)^2$$

$$= -\frac{N}{2}\ln\left(2\pi\right) - \frac{N}{2}\ln\left(\sigma^2\right) - \frac{1}{2\sigma^2}\left\|\mathbf{x} - A\mathbf{s}\right\|^2$$

The ML estimate of *A* is given by:

$$\frac{d\ln L(A)}{dA} = \frac{1}{\sigma^2}\sum_{n=0}^{N-1} s[n]\left(x[n] - A(-1)^n\right) = 0 \quad \Rightarrow \quad \hat{A}_{ML} = \frac{\sum_{n=0}^{N-1}(-1)^n x[n]}{\sum_{n=0}^{N-1} s^2[n]} = \frac{1}{N}\sum_{n=0}^{N-1}(-1)^n x[n] = \frac{1}{N}\mathbf{s}^T\mathbf{x}$$

Since the estimate is a linear function of the data (in practice, it is the output of the linear time-invariant matched filter), which are Gaussian distributed, the estimate is also Gaussian distributed.

The mean value and variance calculated below.

$$E\left\{\hat{A}_{ML}\right\}=\frac{1}{N}\sum_{n=0}^{N-1}(-1)^n E\left\{x[n]\right\}=\frac{1}{N}\sum_{n=0}^{N-1}(-1)^n A(-1)^n = A\cdot\frac{1}{N}\sum_{n=0}^{N-1}(-1)^{2n} = A$$

Therefore, the estimator is **unbiased**.

$$MSE\left\{\hat{A}_{ML}\right\}=\mathrm{var}\left\{\hat{A}_{ML}\right\}=\mathrm{var}\left\{\frac{1}{N}\sum_{n=0}^{N-1}(-1)^n x[n]\right\}=\frac{1}{N^2}\sum_{n=0}^{N-1}(-1)^{2n}\,\mathrm{var}\left\{x[n]\right\}=\frac{\sigma^2}{N}$$

Therefore, the estimator is **consistent**.

The CRB can be derived as follows:

$$\frac{d^2\ln L(A)}{dA^2}=\frac{d}{dA}\left(\frac{1}{\sigma^2}\sum_{n=0}^{N-1}(-1)^n\left(x[n]-A(-1)^n\right)\right)=-\frac{1}{\sigma^2}\sum_{n=0}^{N-1}(-1)^{2n}=-\frac{N}{\sigma^2}$$

$$CRB(A)=\frac{1}{-E\left\{\dfrac{d^2\ln f_X(\mathbf{x})}{dA^2}\right\}}=\frac{\sigma^2}{N}=MSE\left\{\hat{A}_{ML}\right\}$$

Therefore, the estimator is **efficient**.

(3) Under the assumption that σ^2 is **unknown**, the joint ML estimators of A and σ^2 are derived as follows. The ML estimate of A is the same as before because it does not requires a-priori knowledge of the noise variance σ^2:

$$\hat{A}_{ML}=\frac{1}{N}\sum_{n=0}^{N-1}(-1)^n x[n]$$

Therefore, it has the same structure and properties as before

The ML estimate of σ^2 is obtained by minimizing with respect to σ^2 the "compressed" log-likelihood function $\ln L_c(\sigma^2)$, which is obtained by inserting the ML estimate of A in place of A into the log-likelihood function $\ln L(A,\sigma^2)$:

$$\ln L_c(\sigma^2)=\ln L(\hat{A}_{ML},\sigma^2)=-\frac{N}{2}\ln\left(2\pi\right)-\frac{N}{2}\ln\left(\sigma^2\right)-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\left(x[n]-\hat{A}_{ML}(-1)^n\right)^2$$

$$\frac{d\ln L_c(\sigma^2)}{d\sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{n=0}^{N-1}\left(x[n] - \hat{A}_{ML}(-1)^n\right)^2 = 0 \quad \Rightarrow \quad \hat{\sigma}^2_{ML} = \frac{1}{N}\sum_{n=0}^{N-1}\left(x[n] - \hat{A}_{ML}(-1)^n\right)^2$$

$$E\left\{\hat{\sigma}^2_{ML}\right\} = \frac{1}{N}\sum_{n=0}^{N-1} E\left\{\left(x[n] - \hat{A}_{ML}(-1)^n\right)^2\right\}$$

$$E\left\{\left(x[n] - \hat{A}_{ML}(-1)^n\right)^2\right\} = E\left\{x^2[n]\right\} + E\left\{\left(\hat{A}_{ML}\right)^2(-1)^{2n}\right\} - 2(-1)^n E\left\{x[n]\hat{A}_{ML}\right\}$$

$$E\left\{x^2[n]\right\} = \left(E\left\{x[n]\right\}\right)^2 + \operatorname{var}\left\{x[n]\right\} = A^2(-1)^{2n} + \sigma^2 = A^2 + \sigma^2$$

$$E\left\{\left(\hat{A}_{ML}\right)^2\right\} = \left(E\left\{\hat{A}_{ML}\right\}\right)^2 + \operatorname{var}\left\{\hat{A}_{ML}\right\} = A^2 + \frac{\sigma^2}{N}$$

$$E\left\{x[n]\hat{A}_{ML}\right\} = E\left\{x[n]\hat{A}_{ML}\right\} = E\left\{x[n]\frac{1}{N}\sum_{i=0}^{N-1}(-1)^i x[i]\right\} = \frac{1}{N}\sum_{i=0}^{N-1}(-1)^i E\left\{x[n]x[i]\right\}$$

$$= \frac{1}{N}\sum_{i=0}^{N-1}(-1)^i\left(A^2(-1)^n(-1)^i + \sigma^2\delta[i-n]\right) = A^2(-1)^n + \frac{\sigma^2}{N}(-1)^n = \left(A^2 + \frac{\sigma^2}{N}\right)(-1)^n$$

$$E\left\{\hat{\sigma}^2_{ML}\right\} = \frac{1}{N}\sum_{n=0}^{N-1}\left(A^2(-1)^{2n} + \sigma^2\right) + \frac{1}{N}\sum_{n=0}^{N-1}\left(A^2 + \frac{\sigma^2}{N}\right)(-1)^{2n} - \frac{2}{N}\sum_{n=0}^{N-1}(-1)^n \cdot \left(A^2 + \frac{\sigma^2}{N}\right)(-1)^n$$

$$= \left(A^2 + \sigma^2\right) + \left(A^2 + \frac{\sigma^2}{N}\right) - 2\left(A^2 + \frac{\sigma^2}{N}\right) = \sigma^2 - \frac{\sigma^2}{N} = \sigma^2\left(1 - \frac{1}{N}\right) \neq \sigma^2$$

Thus, the noise variance estimator is **biased**:

$$b\left\{\hat{\sigma}^2_{ML}\right\} = E\left\{\sigma^2 - \hat{\sigma}^2_{ML}\right\} = \sigma^2 - \sigma^2\left(1 - \frac{1}{N}\right) = \frac{\sigma^2}{N},$$

but **asymptotically unbiased**:

$$\lim_{N\to\infty} b\left\{\hat{\sigma}^2_{ML}\right\} = \lim_{N\to\infty}\frac{\sigma^2}{N} = 0 \quad \left[\text{if } \sigma^2 < +\infty\right]$$